Chapter 7, Problem 34P

Problem

Evaluate the following derivatives:

(a) $\frac{d}{dt}[u(t-1)u(t+1)]$

(b) $\frac{d}{dt}[r(t-6)u(t-2)]$

(c) $\frac{d}{dt}[\sin 4tu(t-3)]$

Step-by-step solution

Step 1 of 6

(a)

Consider the following derivative:

$$y = \frac{d}{dt}\left[u(t-1)u(t+1)\right]$$
$$= u(t-1)\frac{d}{dt}\left[u(t+1)\right] + u(t+1)\frac{d}{dt}\left[u(t-1)\right]$$
$$= u(t-1)\delta(t+1) + u(t+1)\delta(t-1)$$
$$= u(-1-1)\times\delta(t+1) + u(1+1)\times\delta(t-1)$$
$$= u(-2)\delta(t+1) + u(2)\delta(t-1)$$
$$= 0 + 1\times\delta(t-1)$$
$$= \delta(t-1)$$

Therefore, the derivative $\frac{d}{dt}\left[u(t-1)u(t+1)\right]$ is equal to $\boxed{\delta(t-1)}$.

Step 2 of 6

(b)

Consider the following derivative:

$$y = \frac{d}{dt}\left[r(t-6)u(t-2)\right]$$
$$= r(t-6)\frac{d}{dt}\left[u(t-2)\right] + u(t-2)\frac{d}{dt}\left[r(t-6)\right]$$
$$= r(t-6)\delta(t-2) + u(t-2)u(t-6)$$
$$= r(2-6) + u(t-2)u(t-6)$$
$$= r(-4) + u(t-2)u(t-6)$$
$$= 0 + u(t-2)u(t-6)$$
$$= u(t-2)u(t-6)$$

Step 3 of 6

Plot the graph of $u(t-2)$ as shown in figure 1.

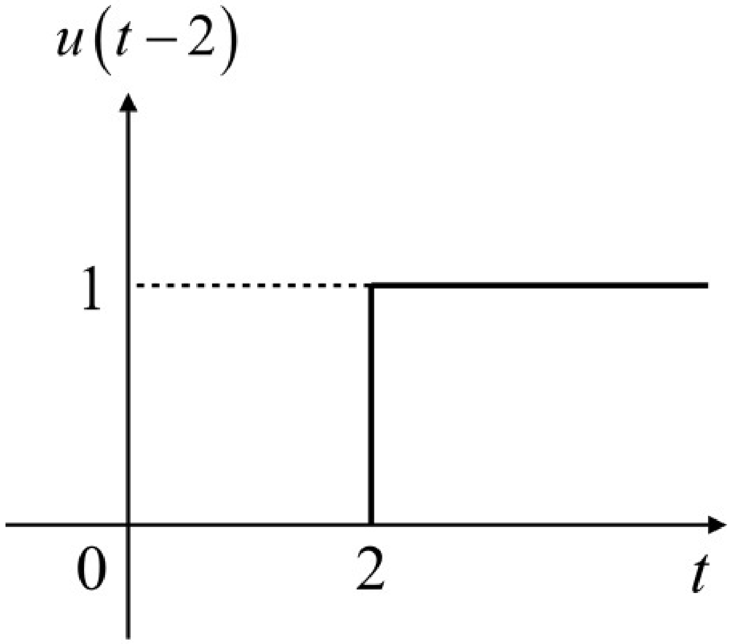

Figure 1

Step 4 of 6

Plot the graph of $u(t-6)$ as shown in figure 2.

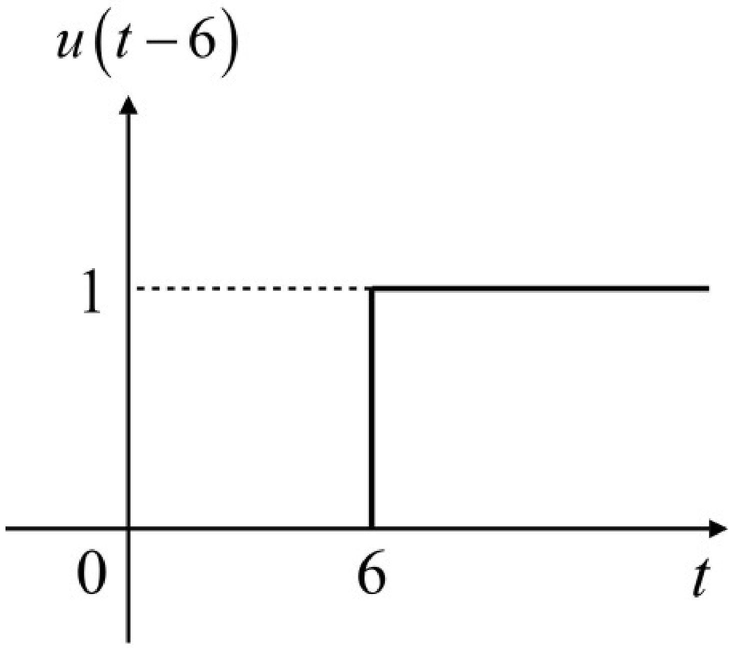

Figure 2

Step 5 of 6

Multiply the signals shown in figure 1 and figure 2.

$$y = u(t-2)u(t-6)$$
$$= u(t-6)$$

Therefore, the derivative $\frac{d}{dt}\left[r(t-6)u(t-2)\right]$ is equal to $\boxed{u(t-6)}$.

Step 6 of 6

(c)

Consider the following derivative:

$$y = \frac{d}{dt}\left[(\sin 4t)u(t-3)\right]$$
$$= \sin 4t\frac{d}{dt}\left[u(t-3)\right] + u(t-3)\frac{d}{dt}[\sin 4t]$$
$$= (\sin 4t)\delta(t-3) + u(t-3)(4\cos 4t)$$
$$= \sin(4\times 3)\delta(t-3) + 4(\cos 4t)u(t-3)$$
$$= \sin(12)\delta(t-3) + 4(\cos 4t)u(t-3)$$
$$= \left[0.2079\times\delta(t-3) + 4(\cos 4t)u(t-3)\right]$$

Therefore, the derivative $\frac{d}{dt}\left[(\sin 4t)u(t-3)\right]$ is equal to,

$\boxed{\left[0.2079\times\delta(t-3) + 4(\cos 4t)u(t-3)\right]}$.